

PROGRAM 1:

```
#include <stdio.h>
void main (){
    printf( "%d\n", 455 );
    printf( "%i\n", 455 );
    /* i sam*/
    printf( "%d\n", +455 );
    printf( "%d\n", -455 );
    printf( "%hd\n", 32000 );
    printf( "%ld\n", 2000000000L );
    printf( "%o\n", 455 );
    printf( "%u\n", 455 );
    printf( "%u\n", -455 );
    printf( "%x\n", 455 );
    printf( "%x\n", 455 );
}
/* indicates successful termination */
/* end main */
```

OUTPUT:

```
Last login: Sun Oct 12 23:08:49 on ttys017
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/1 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/1 ; exit;
455
455
455
-455
32000
2000000000
707
455
4294966841
1c7
1c7

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 2:

```
#include <stdio.h>
int main ( void ){
printf( "%e\n", 1234567.89 );
printf( "%e\n", +1234567.89 );
printf( "%e\n", -1234567.89 );
printf( "%E\n", 1234567.89 );
printf( "%F\n", 1234567.89 );
printf( "%g\n", 1234567.89 );
printf( "%G\n", 1234567.89 );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:10:21 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/2 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/2 ; exit;
1.234568e+06
1.234568e+06
-1.234568e+06
1.234568E+06
1234567.890000
1.23457e+06
1.23457E+06

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```



PROGRAM 3:

```
#include <stdio.h>
int main ( void )
{
    char character = 'A';
    char string[] = "This is a string";
    const char *stringPtr = "This is also a string";
    printf( "%c\n", character );
    printf( "%s\n", "This is a string" );
    printf( "%s\n", string );
    printf( "%s\n", stringPtr);
    return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:18:34 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/3 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/3 ; exit;
A
This isa string
This is a string
This is also a string

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 4:

```
#include <stdio.h>
int main ( void ){
int *ptr;

int x = 12345;
int y;

ptr = &x;

printf( "The value of ptr is %p\n", ptr );
printf( "The address of x is %p\n\n", &x );
printf( "Total characters printed on this line:%n", &y );
printf( " %d\n\n", y );
y = printf( "This line has 28 characters)\n%n" );
printf( "%d characters were printed\n\n", y );
printf( "printing a %% in a format control string\n" );
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:18:55 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/4 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/4 ; exit;
The value of ptr is 0x16ced31c4
The address of x is 0x16ced31c4

Total characters printed on this line: 38

zsh: segmentation fault /Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\
PC/Progs/4

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 5:

```
#include <stdio.h>
int main ( void ){
printf(" %4d\n",1);
printf(" %4d\n",12);
printf(" %4d\n",123);
printf(" %4d\n",1234);
printf(" %4d\n",12345);

printf(" %4d\n",-1);
printf(" %4d\n",-12);
printf(" %4d\n",-123);
printf(" %4d\n",-1234);
printf(" %4d\n",-12345);

return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:19:13 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/5 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/5 ; exit;
  1
 12
123
1234
12345
 -1
-12
-123
-1234
-12345

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 6:

```
#include <stdio.h>
int main( void ){
int i = 873;
double f = 123.94536;

char s[] = "Happy Birthday";
printf( "Using precision for integers\n" );
printf( "\t%.4d\n\t%.9d\n\n", i, i );
printf( "Using precision for floating-point number ");
printf( "\t%.3f\n\t%.3e\n\t%.3g\n\n", f, f, f );
printf( "Using precision for strings\n" );
printf( "\t%.11s\n", s );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:19:31 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/6 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/6 ; exit;
Using precision for integers
    0873
   000000873

Using precision for floating-point number      123.945
    1.239e+02
    124

Using precision for strings
    Happy Birth

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 7:

```
#include <stdio.h>
int main( void ){
printf( "%10s%10d%10c%10F\n\n", "hello", 7, 'a', 1.23 );
printf( "%-105%-10d%-10c%-10f\n", "hello", 7, 'a', 1.23 );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:20:06 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/7 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/7 ; exit;
      hello          7          a  1.230000

%
                                -10d+          0.000000

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 8:

```
#include <stdio.h>
int main( void ){
printf( "%d\n%d\n", 786, -786 );
printf("%d\n%d\n", 786, -786 );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:20:10 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/8 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/8 ; exit;
786
-786
+786
-786

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```


PROGRAM 9:

```
#include <stdio.h>
int main( void ){
printf( "%d\n%d\n", 547, -547 );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:20:45 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/9 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/9 ; exit;
547
-547

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 10:

```
#include <stdio.h>
int main ( void ){
int c = 1427;
double p = 1427.0;
printf( "%#o\n", c );
printf( "%#x\n", c );
printf( "%#X\n", c );
printf( "\n%g\n", p);
printf( "%#g\n", p );

return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:21:05 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/10 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/10 ; exit;
02623
0x593
0X593

1427
1427.00

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 11:

```
#include <stdio.h>
int main( void ){
printf( "%+09d\n", 452 );
printf( "%09d\n", 452 );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:22:31 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/11 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/11 ; exit;
+000000452
000000452

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 12:

```
#include <stdio.h>
int main(void){

    int a;
    int b;
    int c;
    int d;
    int e;
    int f;
    int g;
    printf("Enter seven integers : ");

    scanf( "90%l%l%l%l%l%l%l", &a, &b, &c, &d, &e, &f, &g);
    printf( "The input displayed as decimal integers is:\n" );
    printf( "%d %d %d %d %d %d %d\n", a, b, c, d, e, f, g );
    return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:23:08 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/12 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/12 ; exit;
Enter seven integers : 1 2 3 4 5 6 7
The input displayed as decimal integers is:
118472880 2 118472728 1 1804333672268435455 644893

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 13:

```
#include <stdio.h>
int main(void){
    double a;
    double b;
    double c;
    printf( "enter three floating-point numbers: \n" );
    scanf ( "%le%lf%lg", &a, &b, &c );
    printf( "Here are the numbers entered in plain\n" );
    printf( "floating-point notation:\n" );
    printf( "%f\n%f\n%f\n", a, b, c );
    return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:24:55 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/13 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/13 ; exit;
enter three floating-point numbers:
1.4 0.0 16.7
Here are the numbers entered in plain
floating-point notation:
0.000000
0.000000
0.000000

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 14:

```
#include <stdio.h>
int main( void ){

char x;
char y[ 9 ];

printf( "enter a string: " );
scanf ( "%c%s", &x, y );
printf( "The input was: \n" );
printf( "the character \"%c\" ", x );
printf( "and the string \"%s\" \n", y );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:26:41 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/14 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/14 ; exit;
enter a string: hello
The input was:
the character "h" and the string "ello"

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 15:

```
#include <stdio.h>
int main ( void )
{
    char z[ 9 ];

    printf( "Enter string: " );
    scanf( "%[aeiou]", z );
    printf( "The input was \"%s\\n\"", z );
    return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:29:14 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/15 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/15 ; exit;
Enter string: String
The input was ""

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 16:

```
#include <stdio.h>
int main ( void )
{
    char z[ 9 ];
    printf( "Enter a string: " );
    scanf( "%[Aaeiou]", z );
    printf( "The input was \"%s\\n\"", z );
    return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:31:00 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/16 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/16 ; exit;
Enter a string: string
The input was ""

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```


PROGRAM 17:

```
#include <stdio.h>
int main(void)
{
    int x; int y;
    printf( "Enter a six digit integer: " );
    scanf( "%2d%d", &x, &y );
    printf( "The integers input were %d and %d\n", x, y );
    return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:31:40 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/17 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/17 ; exit;
Enter a six digit integer: 123456
The integers input were 12 and 3456

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

PROGRAM 18:

```
#include <stdio.h>
int main( void ){
int month1;
int day1;
int year1;
int month2;
int day2;
int year2;

printf( "Enter a date in the form mm-dd-yyyy: " );
scanf ( "%d%*c%d%*c%d", &month1, &day1, &year1);
printf( "month = %d day = %d year = %d\n\n",month1,day1,year1);
printf( "Enter a date in the form mm/dd/yyyy: " );
scanf ( "%d%*c%d%*c%d", &month2, &day2, &year2 );
printf( "month = %d day = %d year = %d\n", month2, day2, year2 );
return 0;
}
```

OUTPUT:

```
Last login: Sun Oct 12 23:31:58 on ttys001
/Users/maverick/Documents/f\ College\ Work/f\ Sem\ 1/f\ PC/Progs/18 ; exit;
maverick@Maverick-MacBook-Pro ~ % /Users/maverick/Documents/f\ College\ Work/f\
Sem\ 1/f\ PC/Progs/18 ; exit;
Enter a date in the form mm-dd-yyyy: 02-23-2000
month = 2 day = 23 year = 2000

Enter a date in the form mm/dd/yyyy: 12-12-2012
month = 12 day = 12 year = 2012

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```